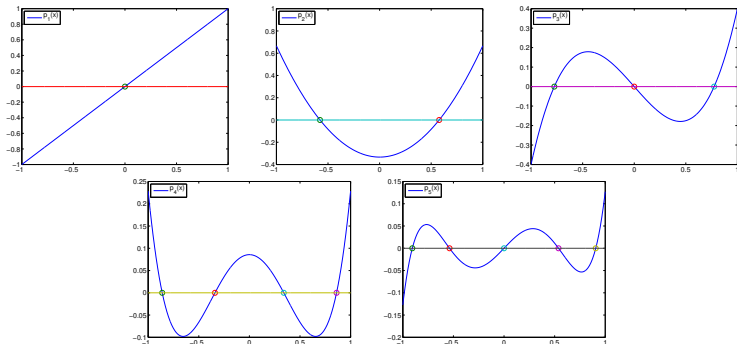


MA378 Chapter 3: Numerical Integration

§3.6 Gaussian Quadrature via Orthogonal Polynomials

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6.0 Today's topics converge to:

- 1 Gaussian Quadrature via Orthogonal Polys
- 2 Precision
 - Polynomial long division
 - The theorem
 - The proof
- 3 Error estimates
- 4 Convergence
- 5 Exercises

6.1 Gaussian Quadrature via Orthogonal Polys

At the start of this section, we introduced the Gaussian Quadrature technique for estimating integrals

$$\int_a^b f(x)dx \approx G_n(f) := \sum_{k=0}^n w_k f(x_k),$$

where the points x_k and weights w_k are chosen to maximise the precision of the method.

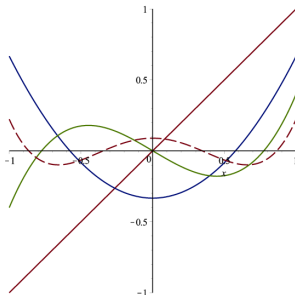
We now have an equivalent way of defining the method, $G_n(\cdot)$, and deriving the coefficients...

6.1 Gaussian Quadrature via Orthogonal Polys

- (a) Construct the set of monic polynomials $\{\tilde{p}_0, \tilde{p}_1, \dots, \tilde{p}_{n+1}\}$ that is orthogonal with respect to the inner product

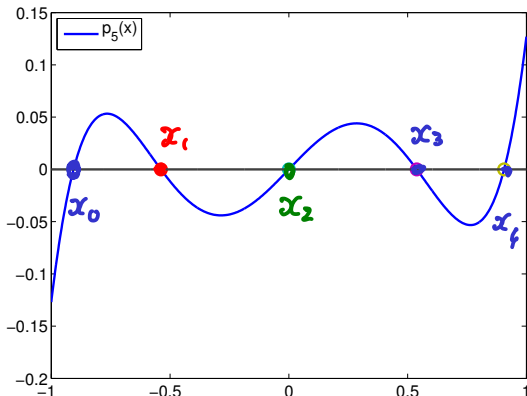
$$(u, v) := \int_a^b u(x)v(x)dx.$$

Note that a and b , the limits of integration for the IP, are the same as for the integral we are trying to estimate.



6.1 Gaussian Quadrature via Orthogonal Polys

- (b) We know that $\tilde{p}_{n+1}$ has $n + 1$ zeros in the interval $[a, b]$. Let these be the quadrature points of the method.
For example, the polynomial $\tilde{p}_5 = x^5 - (10/9)x^3 + (5/21)x$ is shown below, with its zeros highlighted.



6.1 Gaussian Quadrature via Orthogonal Polys

- (c) Take the quadrature weights to be $w_k = \int_a^b L_k(x) dx$, where the L_k are the usual Lagrange polynomials for this set of points.

Recall that if p_n interpolates $f(x)$ at $x_0, x_1, \dots, x_n$ then

$$p_n(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + \dots + f(x_n)L_n(x)$$

is the Lagrange form, where the L_i are the usual Lagrange basis functions.

$$\text{Then } G_n(f) = \int_a^b p_n(x) dx = \sum_{k=0}^n f(x_k) \underbrace{\int_a^b L_k(x) dx}_{w_k}$$

6.1 Gaussian Quadrature via Orthogonal Polys

- (c) Take the quadrature weights to be $w_k = \int_a^b L_k(x) dx$, where the L_k are the usual Lagrange polynomials for this set of points.

Note that this gives that
 G_n has precision at least n .

The following fact will be really useful, which is due to **polynomial long division**. You can read more about it at, for example, https://en.wikipedia.org/wiki/Polynomial_long_division

Polynomial Long Division

Let $f(x)$ and $g(x)$ be polynomials. Then it is always possible to find polynomials $q(x)$ and $r(x)$ such that

$$f(x) = g(x)q(x) + r(x),$$

where the degree of $r(x)$ is less than that of $g(x)$ (or $r \equiv 0$). Here $q(x)$ is called the **quotient**, and $r(x)$ is called the **remainder**

Example: if $f(x) = x^5 + 2x^4 - x^3 + 3x^2 - x + 1$, and $g(x) = x^3 - x + 2$, then $q(x) = x^2 + 2x$, and $r(x) = 3x^2 - 5x + 1$.

The key property of this method is stated in the following theorem.

Theorem 6.1

Let $x_0, \dots, x_n$ be the zeros of $\tilde{p}_{n+1}$, the $(n+1)$ th polynomial in the sequence of orthogonal monic polynomials $\{\tilde{p}_i\}_{i=0}^{\infty}$. Set

$$G_n(f) = w_0 f(x_0) + \cdots + w_n f(x_n)$$

$$\text{where } w_i = \int_a^b L_i(x) dx. \quad (1)$$

Then $G_n(f)$ has precision $2n + 1$.

Proof: Obviously $G_n(\cdot)$ has precision at least n . Suppose that f is a polynomial of degree at most $2n + 1$, then (because of Polynomial Long Division) we can write $f(x)$ as

$$f(x) = \tilde{p}_{n+1}(x)q(x) + r(x) \quad \deg(q), \deg(r) \leq n.$$

Note that $\tilde{p}_{n+1}(x)$ is zero at $x = x_i \dots$

$$G_n(f) = G_n(\tilde{p}_{n+1}(x)q(x) + r(x))$$

$$= G_n(\tilde{p}_{n+1}(x)q(x)) + G_n(r(x))$$

$$\text{But } G_n(\tilde{p}_{n+1}(x)q(x)) = \sum_{k=0}^n \omega_k \underbrace{\tilde{p}_{n+1}(x_k)}_0 q(x_k)$$

$$\text{So } G_n(f) = G_n(r) = \int_a^b r(x) d\omega$$

Next

$$\int_a^b f(x) dx = \int_a^b \tilde{p}_{n+1}(x) q(x) dx + \int_a^b r(x) dx$$

Since $\deg(q) < \deg(\tilde{p}_{n+1})$ so they are orthogonal:

$$\int_a^b \tilde{p}_{n+1}(x) q(x) dx = 0.$$

Thus

$$\int_a^b f(x) dx = \int_a^b r(x) dx = G_n(r) = G_n(f).$$

□

6.3 Error estimates

Our final task associated with numerical integration is to prove that, as $n \rightarrow \infty$, so $G_n(f) \rightarrow \int_a^b f(x)dx$. We won't do this in full detail, but the key ideas will be presented.

We would like to prove the following error estimate, which is closely related to Theorem 1.5.2 (error in Hermitian interpolation):

Theorem 6.2

$$\int_a^b f(x)dx - G_n(f) = \frac{f^{(2n+2)}(\tau)}{(2n+2)!} \int_a^b \pi_{n+1}(x)^2 dx.$$

Idea: show that Gaussian Quadrature is the same as integrating the Hermite interpolant to f , even though $f'(x_k)$ is not involved.

for some $\tau \in [a, b]$. $\pi_{n+1}(x)$: nodal polynomial.

6.3 Error estimates

To do this we need to establish several facts. In each case we make use of Theorem 6.1, a consequence of which is that, if f is a polynomial of degree at most $2n + 1$, then

$$\int_a^b f(x)dx = G_n(f) = \sum_{k=0}^n w_k f(x_k).$$

Recall the basis functions that we use for **Hermite Interpolation**:

$$H_i(x) = [L_i(x)]^2(1 - 2L'_i(x_i)(x - x_i)), \rightarrow H_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

$$K_i(x) = [L_i(x)]^2(x - x_i). \quad K_i(x_k) = 0 \quad \forall i, k.$$

The integrals of these polynomials have surprising properties...

$$\text{where } L_i(x_k) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases}$$

6.3 Error estimates

The key idea here is to exploit that H_i and the K_i are polynomials of degree $2n + 1$, and that $G_n(\cdot)$ has precision $2n + 1$.

$$\blacktriangleright \int_a^b H_i(x) dx = \int_a^b L_i(x) dx.$$

$$\begin{aligned} \int_a^b H_i(x) dx &= G_n(H_i) = \sum_{k=0}^n \omega_k H_i(x_k) \\ &= \omega_i = \int_a^b L_i(x) dx. \end{aligned}$$

$$\blacktriangleright \int_a^b K_i(x) dx = 0.$$

$$\int_a^b K_i(x) dx = G_n(K_i) = \sum_{k=0}^n \omega_k \underbrace{K_i(x_k)}_{=0} = 0.$$

6.3 Error estimates

Now we can deduce the error estimate:

(i) By definition, $G_n(f) = \sum_{j=0}^n w_j f(x_j)$

(ii) By construction, $w_j = \int_a^b L_j(x) dx$. Therefore

$$G_n(f) = \sum_{j=0}^n \left(\int_a^b L_j(x) dx \right) f(x_j)$$

(iii) From the previous slide we now have that $G_n(f) =$

$$\sum_{j=0}^n \left(\int_a^b H_j(x) dx \right) f(x_j) + \sum_{j=0}^n \left(\int_a^b K_j(x) dx \right) f'(x_j)$$

6.3 Error estimates

(iv) Rearranged to get

$$G_n(f) = \int_a^b \sum_{j=0}^n \left(H_j(x) f(x_j) + K_j(x) f'(x_j) \right) dx$$

(v) So $G_n(f) = \int_a^b p_{2n+1}(x) dx$ where p_{2n+1} is the Hermite interpolant to f .

(vi) From Thm 5.2 of Chapter 1:
$$f(x) - p_{2n+1}(x) = \frac{f^{(2n+2)}(\tau)}{(2n+2)!} [\Pi_{n+1}(x)]^2$$

(vii) So
$$\left| \int_a^b f(x) dx - G_n(f) \right| = \frac{f^{(2n+2)}(\tau)}{(2n+2)!} \int_a^b [\Pi_{n+1}(x)]^2 dx$$

6.3 Error estimates

$$[L_k(x_k)]^2 = \sum_{i=0}^n \begin{matrix} 1 & i=k \\ 0 & i \neq k \end{matrix}$$

Next we want to show that each of the w_k are positive. From (1) we have that the Gaussian Quadrature weights are $w_k = \int_a^b L_k(x) dx$ where the L_k are the usual Lagrange Polynomials:

$$L_k(x_j) = \begin{cases} 1 & k = j \\ 0 & k \neq j, \end{cases}$$

associated with the Gaussian interpolation points. Then in fact....

Since $L_k(x)$ is a polynomial of degree n , then $(L_k(x))^2$ is a poly of degree $2n$. So

$$G_n([L_k(x)]^2) = \int_a^b (L_k(x))^2 dx. \text{ But}$$

$$(L_k(x))^2 \geq 0 \quad \forall x. \text{ So}$$

$$0 < \int_a^b (L_k(x))^2 dx = G_n((L_k)^2) = \sum_{i=0}^n w_i [L_k(x_i)]^2 = w_k$$

6.3 Error estimates

So, since

$$w_i = \int_a^b [L_i(x)]^2 dx,$$

we have that all the w_k are all positive. It follows directly that $0 < w_k < (b - a)$, for $k = 0, 1, 2, \dots, n$:

$$b - a = \int_a^b 1 dx = G_n(1) = \sum_{k=0}^n w_k.$$

So, since $w_k > 0 \quad \forall k$ and
 $\sum_{k=0}^n w_k = b - a$, each $w_k < (b - a)$

6.4 Convergence

Section 10.4 of Süli and Mayer's also covers this, though from a different angle. One of the most interesting aspects of this theory is given in Theorem 10.2 of that book:

Theorem 6.3

$$\lim_{n \rightarrow \infty} G_n(f) = \int_a^b f(x) dx.$$

An outline of the proof is given below. Read it if you have time; we won't cover it in class and it will not be on the MA378 exam.

We finished here Wed 18 March. You can read the rest of the slides if interested: they won't be on the exam.

6.4 Convergence

The **Weierstrass approximation theorem**, tells us that, for any $\epsilon > 0$, there exists a polynomial p such that $|f(x) - p(x)| \leq \epsilon$. Let n be the degree of this polynomial. Let $G_n(\cdot)$ be the $n + 1$ point Gaussian Quadrature rule. Then

$$\begin{aligned} \int_a^b f(x)dx - G_n(f) &= \\ \int_a^b f(x) - p(x)dx + \int_a^b p(x)dx - G_n(p) + G_n(p) - G_n(f). \end{aligned}$$

But because $G_n(\cdot)$ is exact for polynomials of degree n , $\int_a^b p(x)dx - G_n(p) = 0$. Using this, and the triangle inequality,

$$\begin{aligned} \left| \int_a^b f(x)dx - G_n(f) \right| &\leq \\ &\left| \int_a^b f(x) - p(x)dx \right| + \left| G_n(p) - G_n(f) \right|. \end{aligned}$$

6.4 Convergence

But

$$\left| \int_a^b f(x) - p(x) dx \right| \leq \int_a^b \epsilon dx = \epsilon(b - a).$$

Also,

$$\begin{aligned} |G_n(f) - G_n(p)| &= |G_n(p - f)| = \\ &= \left| \sum_{k=0}^n w_k (f(x_k) - p(x_k)) \right| = \\ &= \sum_{k=0}^n w_k |f(x_k) - p(x_k)|, \end{aligned}$$

because all the w_i are positive. Then, using that $\sum_k w_k = b - a$ and $|f(x) - p(x)| \leq \epsilon$, we get

$$|G_n(f) - G_n(p)| \leq \epsilon(b - a).$$

6.4 Convergence

Combining these we have shown that for any ϵ , one can always find n such that

$$\left| \int_a^b f(x)dx - G_n(p) \right| \leq 2\epsilon(b-a).$$

.....

One of the key ingredients to this proof was that the w_i are all positive. That is not the case for Newton-Cotes methods—for large enough n their quadrature weights can be negative—so no similar result is possible.

6.5 Exercises

Exercise 6.1

Give a complete proof of Theorem 6.1 (i.e., that $G_n(\cdot)$ has precision $2n + 1$).

Exercise 6.2

Show that it is impossible to choose $n + 1$ quadrature points and weights so that the $n + 1$ -point quadrature rule

$$\int_a^b f(x)dx \approx \sum_{k=0}^n w_k f(x_k)$$

has precision $2n + 2$.

Hint: To show the method does not have precision $2n + 2$, you just need to give a an example of a single polynomial p of degree exactly $2n + 2$ for which

$$\int_a^b p(x)dx \neq \sum_{k=0}^n w_k f(x_k).$$